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Conditional Cash Transfers as Stabilization Tool: a HANK Approach

1. Model

The setup is a standard New Keynesian model with incomplete markets and labor supply risk. Time is discrete and runs from $t = 0$ to infinity. There are four types of agents in the economy: households, firms, labor unions, and the government.

Households are heterogeneous (Aiyagari 1994), workers and firms join in a formal-informal setup, and the rest of the economy follows the textbook New Keynesian model (Galí 2015). There is no aggregate uncertainty and the economy starts in the deterministic steady-state.

Households

Assume there exists a continuum $i \in [0, 1]$ of ex-ante identical households, but experiencing idiosyncratic shocks in labor productivity and discount factor.

Households enjoy a CRRA-GHH utility over consumption $c_{i,t}$ and worked hours $h_{i,t}$ (Greenwood, Hercowitz, and Huffman 1988)[1], which we define $u(\tilde{c}_{i,t}) = u(c_{i,t} - v(h_{i,t}))$ as

$$\max_{c_{i,t}, h_{i,t}} \ \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{\tilde{c}_{i,t}^{1-\frac{1}{\gamma}}}{1 - \frac{1}{\gamma}} \right], \quad \text{with } v(h) = \psi \frac{h^{1+\frac{1}{\varphi}}}{1 + \frac{1}{\varphi}}$$

where γ is the elasticity of intertemporal substitution (EIS) and φ is the Frisch elasticity of labor supply.

A household faces idiosyncratic labor productivity shocks $e_{i,t}$ with $\mathbb{E}[e_{i,t}] = 1$, and discount factor heterogeneity $\beta_i \in \{\beta_L, \beta_H\}$, and we fix ω_I impatient households. They can work in the formal or informal sector, or be unemployed $s \in \{F, I, U\}$. Each sector has a specific productivity $\theta^s_{i,t}$ which their wage depends on, re-drawn on each new offer.

Formal workers earn labor income $(1 - \tau_\ell)\theta^F_{i,t} w_t e_{i,t} h_{i,t}$, while informal workers earn $\theta^I_{i,t} w_t e_{i,t} h_{i,t}$, where w_t is the baseline wage, τ_ℓ is a fixed labor income tax rate. Let $y_t(s, e, \theta)$

[1] The GHH specification assumes zero income effect, so that the hours decision is separable from the consumption-savings problem.

be this non-financial income,

$$y_t(s,e,\theta) = \begin{cases} (1-\tau_\ell)\theta^F w_t h_t e, & \text{if formal;} \\ \theta^I w_t h_t e + T_t \cdot \mathbb{1}[\text{eligible}], & \text{if informal;} \\ T_t, & \text{if unemployed.} \end{cases}$$

We assume that formal wages are observable by the government, so that only informal workers and unemployed may be eligible to receive *Bolsa Família* (BF). We assume that informal households are eligible if $y_{i,t}(I,e,\theta) < \bar{y}$ and that the total number of beneficiaries is BF_t.

Households save into liquid assets $a_{i,t}$ with ex-ante real return $1+r_t = (1+i_{t-1})/(1+\pi_t)$, where i_t represents the nominal interest rate and π_t denotes inflation. They also receive lump-sum transfers τ_t by the government and dividends $d_{i,t}$ from firms, which we assume to be proportional to productivity $d_{i,t} = d_t \cdot e_{i,t}$[2]. Their budget constraint at time t reads:

$$c_{i,t} + a_{i,t} = (1+r_t)a_{i,t-1} + y_{i,t}(s,e,\theta) + d_{i,t} + \tau_t.$$

Incomplete markets impose a borrowing constraint $a_{i,t} \geq \underline{a}$. Frictions in the labor market imply formal households delegate their labor supply decisions to labor unions, and thus take worked hours as given. While, informal ones choose hours to maximize their utility, thus, for each i,

$$\psi\,(h)^{1/\varphi} = \theta w e \quad \Leftrightarrow \quad h^I(e) = \left[\frac{\theta w e}{\psi}\right]^{\varphi}.$$

This generates a productivity cutoff for informal workers of

$$\theta w e \left(\frac{\theta w e}{\psi}\right)^{\varphi} < \bar{y} \quad \Leftrightarrow \quad e < e^* = \frac{(\psi^{\varphi}\bar{y})^{\frac{1}{1+\varphi}}}{\theta w}.$$

Such as Esteban-Pretel and Kitao (2021) and Finamor (2025), given a sector s, we assume that each worker faces an exogenous layoff probability δ_s, and receives an offer from s with probability π_s. The worker accepts the offer with the higher value, smoothed by additive idiosyncratic preference shocks explained later.

[2]Dividends are proportional to productivity in order to increase inequality. We aim to perform robustness tests with other specifications.

The Bellman Equation for each individual i in sector $s \in \{F, I\}$ is

$$
\begin{aligned}
V_t^s(a, e, \beta, \theta) = \max_{\tilde{c}, a'} \Bigg\{ \frac{\tilde{c}^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta \mathbb{E}_t \Big[\delta_s V_{t+1}^U(.) + (1-\delta_s) \Big(& \\
& \pi_F (1-\pi_I) \max\Big\{ V_{t+1}^s(., \theta), V_{t+1}^F(., \theta') \Big\} \\
& + (1-\pi_F)\pi_I \max\Big\{ V_{t+1}^s(., \theta), V_{t+1}^I(., \theta') \Big\} \\
& + \pi_F \pi_I \max\Big\{ V_{t+1}^s(., \theta), V_{t+1}^I(., \theta'), V_{t+1}^F(., \theta') \Big\} \\
& + (1-\pi_F)(1-\pi_I) V_{t+1}^s(., \theta) \Big) \Big] \Bigg\}
\end{aligned}
$$

$$
\begin{aligned}
\text{s.t.} \qquad & c + a' = (1 + r_t)a + y_t(s, e, \theta) + d_{i,t} + \tau_t \\
& a' \in [\underline{a}, \infty), \qquad \tilde{c} = c - v(h),
\end{aligned}
$$

and for each unemployed worker is

$$
\begin{aligned}
V_t^U(a, \beta) = \max_{c, a'} \Bigg\{ \frac{c^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta \mathbb{E}_t \Big[& (1-\pi_F)(1-\pi_I) V_{t+1}^U(.) \\
& + \pi_F (1-\pi_I) \max\Big\{ V_{t+1}^U(.), V_{t+1}^F(.) \Big\} \\
& + (1-\pi_F)\pi_I \max\Big\{ V_{t+1}^U(.), V_{t+1}^I(.) \Big\} \\
& + \pi_F \pi_I \max\Big\{ V_{t+1}^U(.), V_{t+1}^I(.), V_{t+1}^F(.) \Big\} \Big] \Bigg\}
\end{aligned}
$$

$$
\begin{aligned}
\text{s.t.} \qquad & c + a' = (1 + r_t)a + y_t(U) + d_{i,t} + \tau_t \\
& a' \in [\underline{a}, \infty).
\end{aligned}
$$

Formal workers maximize their utility taking hours as given, so their labor disutility is internalized at the union level, not by individual households. However, informal hours are chosen and therefore included in the Euler Equation. Given their decision, their consumption-equivalent income is

$$
w^I e h_t - v(h_t) = w^I e h_t - \psi \frac{h_{i,t} h_{i,t}^{1/\varphi}}{1 + \frac{1}{\varphi}} = \frac{w^I e h_t}{1 + \varphi}.
$$

As in Caliendo, Dvorkin, and Parro (2019), before deciding, the households receive an offer $A \in \{F, I, FI\}$ and draws a vector $\{\varepsilon^j\}$ over the options $j \in \mathcal{S}$, with scale σ_S[3], and picks the option j maximizing $V^j + \varepsilon^j$. This delivers a closed-form expected value,

$$
\mathbb{P}^A(j) = \frac{\exp(V^j/\sigma_S)}{\sum_{k \in \mathcal{S}} \exp(V^k/\sigma_S)}, \qquad \mathbb{E}\Big[\max_{j \in \mathcal{S}} \Big\{ V^j + \varepsilon^j \Big\}\Big] = \sigma_S \ln \sum_{j \in \mathcal{S}} \exp\Big(V^j/\sigma_S\Big)\ ^4.
$$

[3] σ_S governs the smoothness of the decision. As $\sigma_S \to 0$ it collapses to the deterministic max.
[4] We use a standard linear weighted approximation for the log-exp sum.

These shocks render the transition a differentiable function, which allows it to be linearized when computing the sequence-space Jacobians, as in Iskhakov et al. (2017). Therefore, the sector transition matrix $\Pi_t^s = \mathbb{P}(s'|s)$ is given by

$$\Pi_t^s = \begin{bmatrix} (1-\delta_F)\phi_F(F) & (1-\delta_F)\phi_F(I) & \delta_F \\ (1-\delta_I)\phi_I(F) & (1-\delta_I)\phi_I(I) & \delta_I \\ \phi_U(F) & \phi_U(I) & \phi_U(U) \end{bmatrix},$$

where $\phi_s(s')$ is the probability of accepting an offer s' given that the worker is in sector s,

$$\begin{aligned} \phi_F(I) &= \pi_F \pi_I \, \mathbb{P}'^{FI}(I) + (1-\pi_F)\pi_I \, \mathbb{P}^I(I), & \phi_F(F) &= 1 - \phi_F(I), \\ \phi_I(F) &= \pi_F \pi_I \, \mathbb{P}^{FI}(F) + \pi_F(1-\pi_I) \, \mathbb{P}^F(F), & \phi_I(I) &= 1 - \phi_I(F), \\ \phi_U(F) &= \pi_F \pi_I \, \mathbb{P}^{FI}(F) + \pi_F(1-\pi_I) \, \mathbb{P}^F(F), & & \\ \phi_U(I) &= \pi_F \pi_I \, \mathbb{P}^{FI}(I) + (1-\pi_F)\pi_I \, \mathbb{P}^I(I), & \phi_U(U) &= 1 - \phi_U(F) - \phi_U(I). \end{aligned}$$

We denominate the steady-state size of each sector $s \in \{F, I, U\}$ as α_s, such that $\alpha_F + \alpha_I + \alpha_U = 1$. Let $\Lambda_t(s, \beta, e, a)$ be the household distribution over a given period of time.

Firms

In the firm side, there is a representative final-good producer that aggregates a continuum of $j \in [0, 1]$ intermediate inputs produced by formal firms,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\varepsilon > 0$ is a constant substitution elasticity (CES).

Each intermediate good j is produced by a monopolistic competitive firm $y_{j,t} = Z_t l_{j,t}$. They hire labor at real wage w_t and set a constant markup of $\mu = \frac{\varepsilon}{\varepsilon-1}$. Hence, $P_t = \mu Z_t W_t$, and real wage equals $w_t = Z_t/\mu$ at every date in steady-state.

We assume sticky prices following Rotemberg (1982), standard in Neo-Keynesian's literature, with a quadratic adjustment cost

$$\Psi_t(\pi_t) = \frac{\mu}{\mu - 1} \frac{1}{2\kappa} \left[\log(1+\pi_t)\right]^2 Y_t.$$

Firms' dividends are taxed at the same rate τ_ℓ as labor income, so $d_t = (1-\tau_\ell)(Y_t - w_t L_t) - \Psi_t(\pi_t)$, where $L_t = \mathbb{E}[l_{j,t}]$ is the aggregate formal labor. Such as Hagedorn, Manovskii, and Mitman (2019), the Phillips curve for price inflation is given by

$$\ln(1+\pi_t) = \kappa \left[\frac{w_t}{Z_t} - \frac{1}{\mu} \right] + \frac{\ln(1+\pi_{t+1})}{1+r_{t+1}} \cdot \frac{Y_{t+1}}{Y_t},$$

Additionally, there is a representative informal firm operating in a perfectly competitive market and producing under constant returns to scale, so $Y_t^I = w_t L_t^I$.

Unions

Nominal wages are assumed to be sticky as in Erceg, Henderson, and Levin (2000), where unions set nominal wages to maximize agent utility subject to Rotemberg (1982)'s adjustment costs. We assume that unions allocate all formal labor hours uniformly across agents, so $h_{i,t} = h_t$ (Auclert, Rognlie, and Straub 2025).

Also, as in Hagedorn, Manovskii, and Mitman (2019), union's objective is to maximize the utility of an agent with average consumption $\bar{C}_t = \int_0^1 c_{it}\,di$, discounted by the average discount factor $\bar{\beta} = \lambda_I \beta_L + (1-\lambda_I)\beta_H$ (Auclert, Rognlie, Souchier, et al. 2021). This results in a Phillips curve for wage inflation $\pi_t^w = (1+\pi_t)w_t/w_{t-1}$ given by

$$\ln(1+\pi_t^w) = \kappa_w \left[\psi \cdot h_t^{1/\varphi} \cdot \tilde{C}_t^{1/\gamma} - \frac{(1-\tau_\ell)w_t L_t}{\mu_w}\right] + \bar{\beta}\ln(1+\pi_{t+1}^w).$$

The Phillips Curve calculations for price and wage are found in the Appendix.

Government

Monetary Policy follows a standard lagged Taylor Rule, without microfoundations and only with an inflation target, $i_t = r_{t-1}^* + \phi\pi_{t-1}$, where r^* is the neutral real interest rate, and $\phi > 1$ (Taylor 1993).

In the governmental sphere, we have a government that collects taxes, makes transfers, and incurs debt; therefore, the government's budget constraint is characterized by

$$\tau_t + T_t \cdot \mathrm{BF}_t + (1+r_t)B_{t-1} = B_t + \tau_\ell Y_t.$$

We consider two types of government policy, a revenue-neutral and a deficit-financed. The policymaker chooses a path of the conditional transfers and adjusts transfers or debt such that the budget is balanced at each period.

The revenue-neutral policy sets a path of T_t and adjusts transfers such that

$$\tau_t = \tau_\ell Y_t - T_t \cdot \mathrm{BF}_t - r_t\bar{B},$$

which means that the fiscal policy is passive and the monetary policy is active. The nominal interest rate adjust at least one-for-one to changes in inflation.

However, governments normally finance themselves by increasing deficit and issuing new debt. The deficit-financed policy sets a path of T_t and τ_t and adjusts debt, keeping the nominal interest rate fixed $i_t = \bar{i}$. We assume that the government repays its debt in the long run by reducing transfer (when $\tau_t > 0$) or increasing taxes (when $\tau_t < 0$), with elasticity ϕ_B,

$$\tau_t = \bar{\tau} - \phi_B \cdot (B_{t-1} - \bar{B}).$$

This situation leads to larger multipliers and great inequality (Auclert, Rognlie, and Straub 2023). In the presence of nominal rigidity and non- Ricardian consumer behavior, Angeletos, Lian, and Wolf (2024) show that fiscal deficits of this sort self finance, so $\lim_{t\to\infty} B_t = \bar{B}$.

Market Clearing

Finally, the asset, labor, and goods markets reach equilibrium (Market Clearing Conditions). First, the asset market reaches equilibrium when household savings equal government bonds

$$A_t = \int a_{i,t} \, d\Lambda_t(s, \beta, e, a) = B_t;$$

the labor market clears when the share of employment in the workforce equals the firm's demand for labor

$$L_t = \int \mathbb{1}\{s_{i,t} = F\} \cdot \theta^F_{i,t} \cdot e_{i,t} \cdot h_{i,t} \, d\Lambda_t(s, \beta, e, a);$$
$$L^I_t = \int \mathbb{1}\{s_{i,t} = I\} \cdot \theta^I_{i,t} \cdot e_{i,t} \cdot h_{i,t} \, d\Lambda_t(s, \beta, e, a);$$

and the goods market clear when $Y + Y^I = C + \Psi_t$. We set the goods market to clear by Walras' Law. Lastly, the BF benefit is given to all eligible households,

$$\mathrm{BF}_t = U_t + \int \mathbb{1}\{s_{i,t} = I, e_{i,t} < e^*\} \, d\Lambda(s, \beta, e, a).$$

2. Calibration

Data

To calibrate this model, we use Brazilian data and standard parameters in HANK Literature. Most of the data refers to their average values in 2025.

The labor market moments were obtained in *Pesquisa Nacional por Amostra de Domicílio Contínua* (PNAD-C), an annual survey reporting formality status, wages, weekly worked hours, etc. This panel is interesting to match the sector transitions and the worked hours and wages, for both formal and informal households. However, since it is a survey, it might have under-reporting of informal households.

Some general macroeconomic measures were also used, such as the Government Debt (in the Brazilian National Treasure), the interest rates, and the inflation, available in the *Índice Brasileiro de Geografia e Estatística* (IBGE). The Gini coefficient of wealth was obtained from the 2025 Global Wealth Report, from UBS. *Bolsa Família* data is available in *Cadastro Único* (CadÚnico), in VIS Data, a platform with several statistics about the benefit.

Additionally, Annual PNAD-C has information about whether a individual is beneficiary of the benefit. We identify each household and consider every individual living in a beneficiary household as receiving the benefit. Among those, we calculate the mean reported benefit T and the mean share of beneficiaries.

Table 1: Bolsa Família 2025: Statistics

	PNAD	VisData	Ratio
Households (M)	13.64	18.71	1.37
Individuals (M)	46.60	48.92	1.05
People/HH	3.42	2.61	0.77
Value (R$/month)	669.55	710.65	1.06

Table 2: Bolsa Família 2025: Coverage

	Coverage
BF / Labor Force	0.161
BF in Formal	0.072
BF in Informal	0.239
BF in Unemployed	0.336
BF in Non-Participating	0.218
$T/\mathbb{E}(y^F)$	0.155

The rule to receive BF is based on a R$ 218 threshold for the family income per capita, therefore there are some formal individuals receiving the benefit. Also, CadÚnico normally delay to remove the benefit for non-eligible individuals, which can explain why some formal individuals receive BF.

In order to construct quarter-to-quarter transition probabilities across the three sectors $s \in \{F, I, U\}$, we reconstruct the PNAD-C rotating panel using Osório (2022), which estimates the dwelling and individual identifiers in the survey. The estimation of the pandemic years is not reliable.

The formality status is defined as (i) formal workers have signed work card, are militaries, or employers, and (ii) informal workers do not have a signed work card, are self-employed, or unpaid family. We only consider household in the labor force, dropping those who are non-participating, retired, or students, to keep the empirical states aligned with the model's.

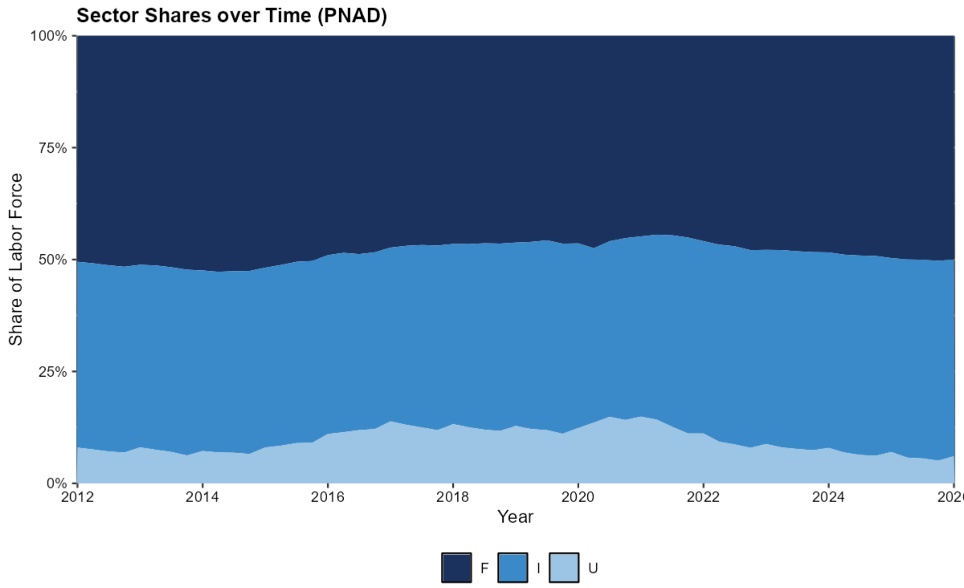

Figure 1: Share of Formal, Informal, and Unemployed Households

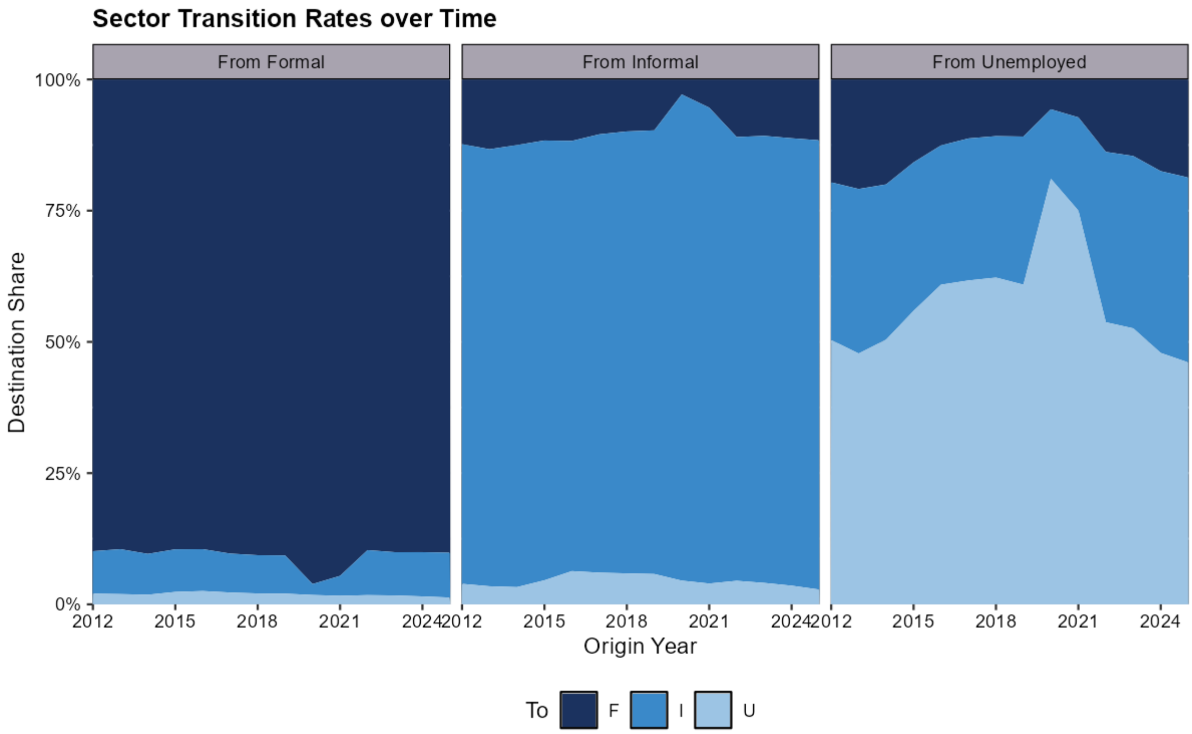

Figure 2: Sector Transitions over time

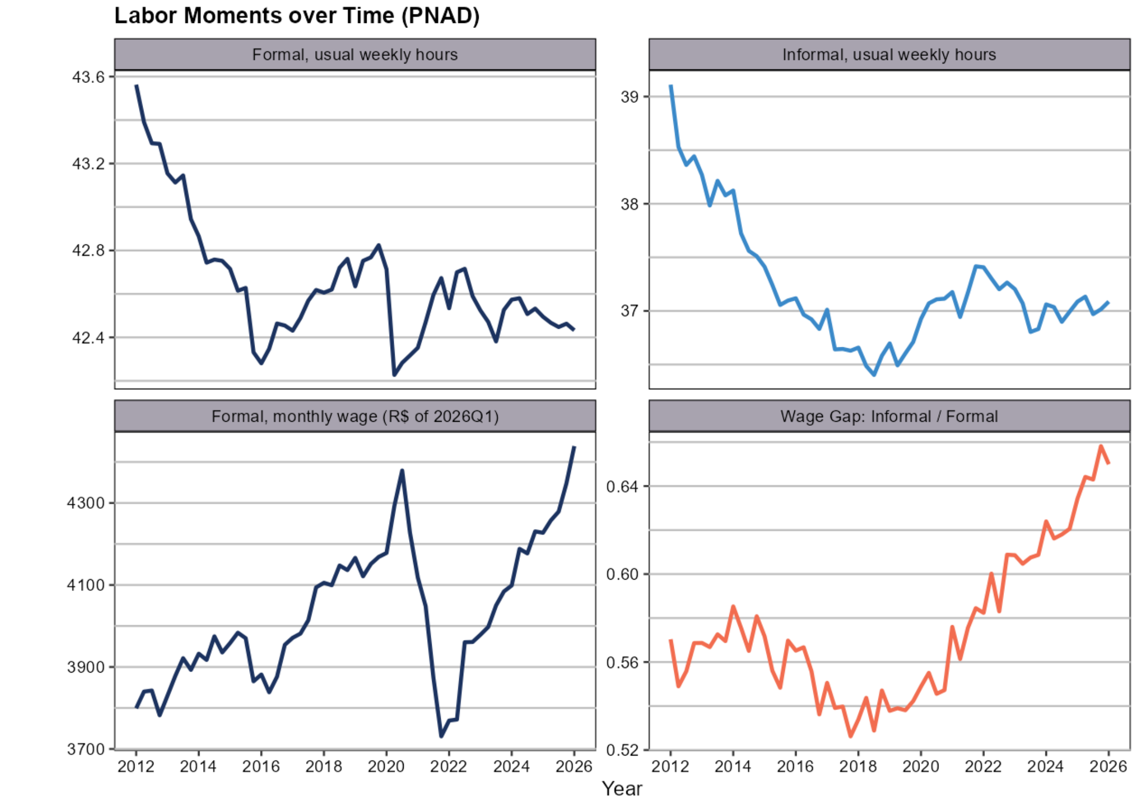

Figure 3: Wage and Worked Hours for each Sector

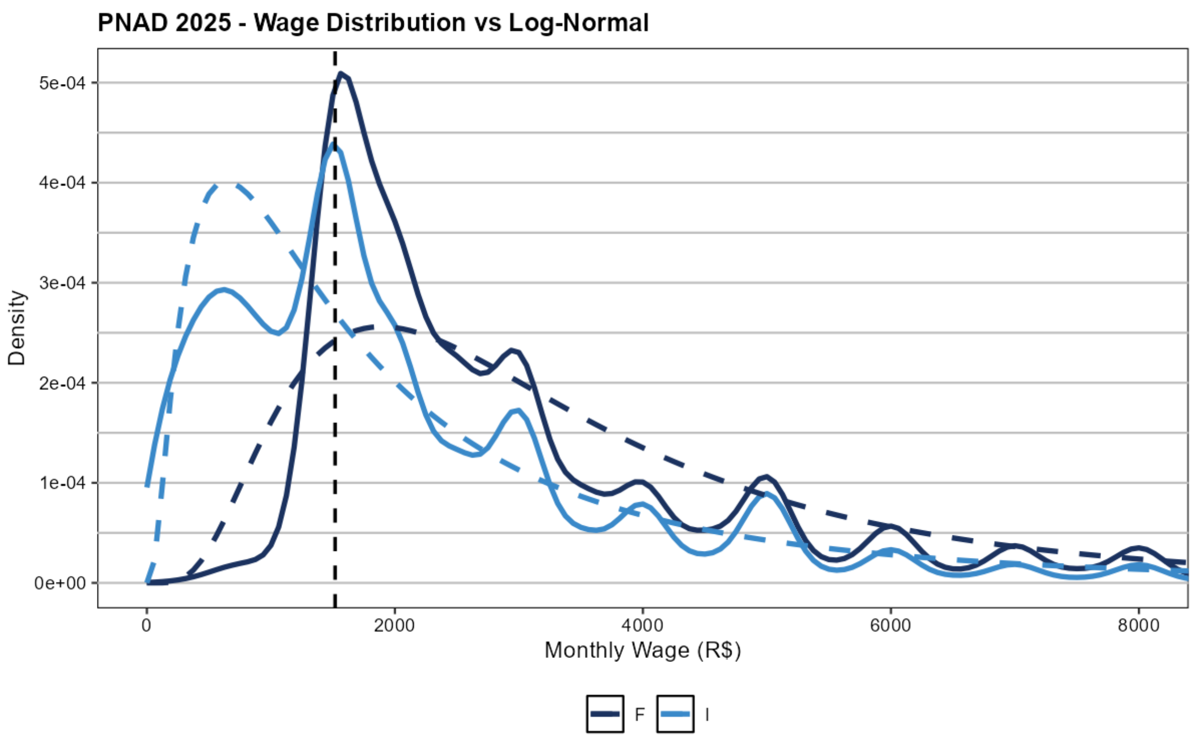

Figure 4: Wage Distribution for each Sector (vs MLE-estimated Log-Normal)

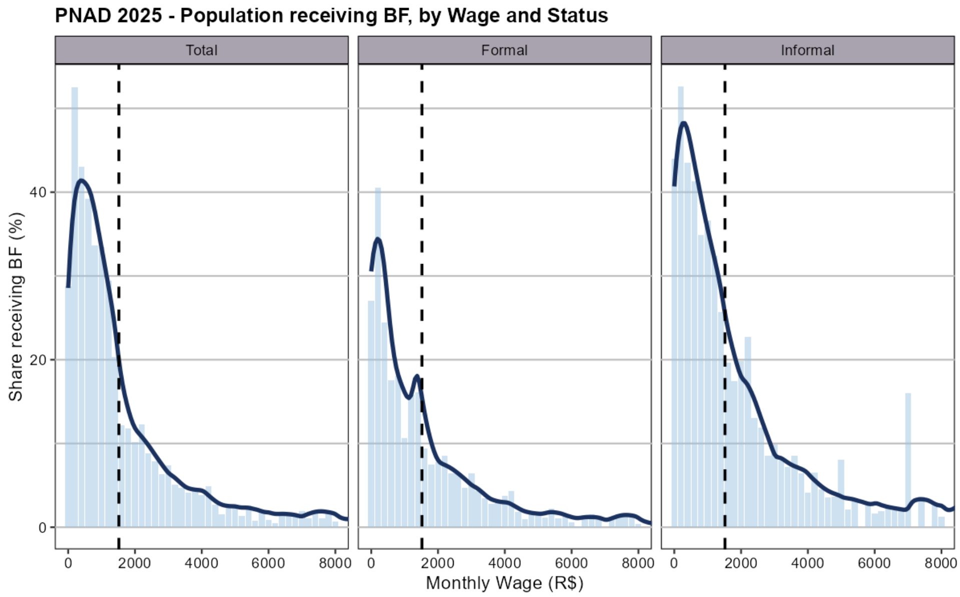

Figure 5: Share of Households Receiving Benefit by Wage Level

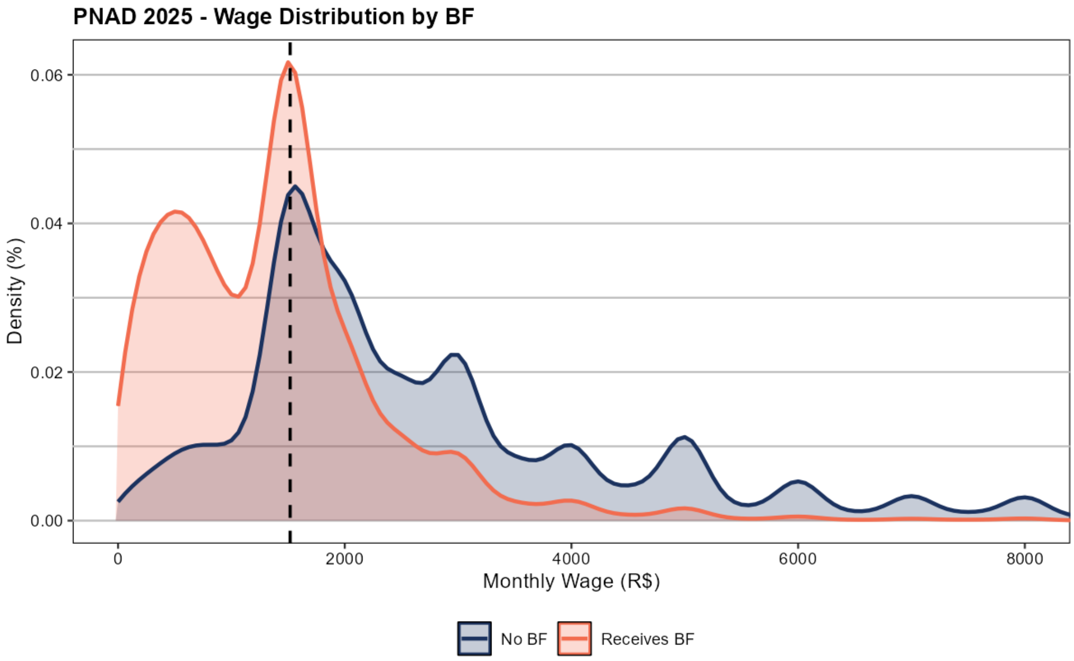

Figure 6: Wage Distribution by Benefit Status

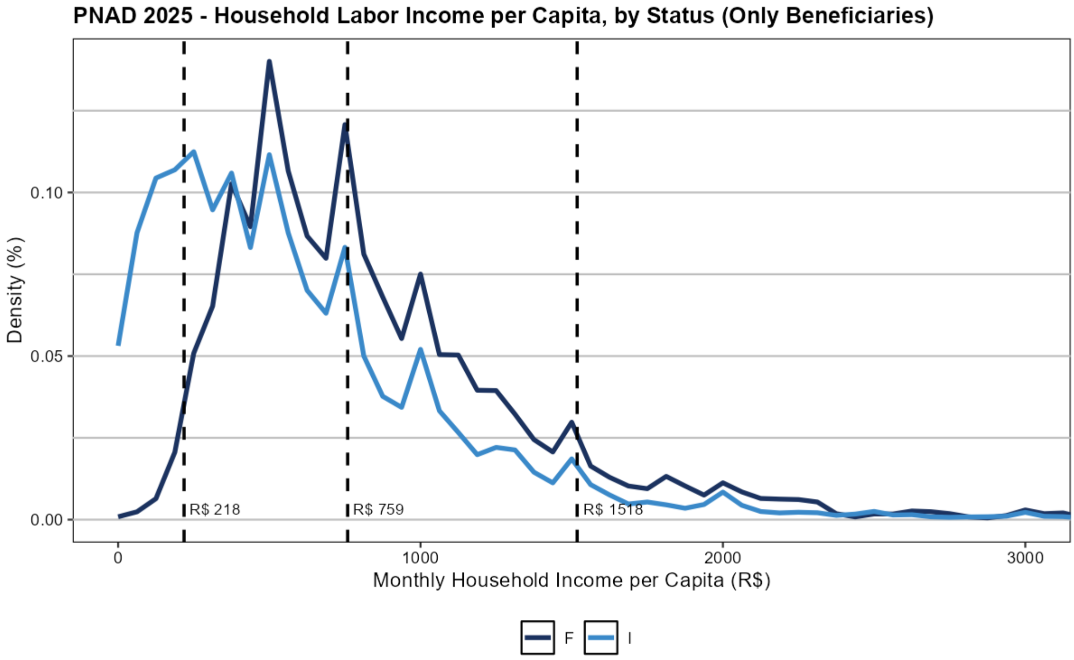

Figure 7: Familiar Wage Distribution by Sector, for Beneficiaries

Methods

The main computational methods are the Sequence-Space Jacobian (SSJ) (Auclert, Bardóczy, et al. 2021), and the Endogenous Grid Method (EGM) (Carroll 2006). The EGM was combined with the discrete choice among sectors using Iskhakov et al. (2017), with taste shocks.

The central idea of SSJ is to summarize the entire model by a matrix equation $\mathbf{F}_t(\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, where $\mathbf{X}$ are the endogenous variables and $\mathbf{Z}$ are the exogenous ones. The Implicit Function Theorem allows us to differentiate this system and obtain General Equilibrium Jacobians, written as

$$d\mathbf{X} = \mathbf{G}\, d\mathbf{Z} = -\mathbf{F}_{\mathbf{X}}^{-1}\mathbf{F}_{\mathbf{Z}}\, d\mathbf{Z}.$$

For each shock in *Bolsa Família*, we can compute the Jacobian $dC = \mathbf{G}_{C,T}\, dT$ of the change in the conditional transfer on household consumption. The Jacobian is computed by perturbing the exogenous variable T and solving the model for the new equilibrium, then calculating the change in consumption C.

Further, we can compute the consumption response to a monetary shock $dC = \mathbf{G}_{C,i}\, di$ in the presence of the transfer, and compare it to the counterfactual. We measure the stabilization effect of the transfer as the difference between the response in the economy with the program and in the $T = 0$ counterfactual,

$$\mathrm{Stab}(T) = dC\Big|_{T>0} - dC\Big|_{T=0},$$

read either as an impulse response or as the induced change in the volatility of consumption $\Delta\mathrm{Var}(C_t)$, a program that smooths fluctuations delivers $\Delta\mathrm{Var}(C_t) < 0$. This effect can be decomposed in a insurance and a composition channel,

$$d\mathbf{X} = d\mathbf{X}^{\text{ins}} + \Big(d\mathbf{X} - d\mathbf{X}^{\text{ins}}\Big) = \mathbf{G}^{\text{ins}}\, d\mathbf{Z} + \Big(\mathbf{G} - \mathbf{G}^{\text{ins}}\Big)\, d\mathbf{Z}.$$

The sectoral block of the transition matrix Π_t is endogenous and depends on the continuation values, $\Pi_t = \Pi(\mathbf{V}_t, \Lambda_t)$, where $\mathbf{V}_t = (V_t^F, V_t^I, V_t^U)$. The household's distribution evolves by combining the asset policy with the Markov transition over the states, $\Lambda_{t+1} = \mathcal{T}(\Pi_t, a_t(s, \theta, \beta, e))$. To solve the steady-state distribution, we iterate until Π_t converges to its fixed point, analogue to Aguirregabiria and Mira (2007).

The insurance channel is captured by the frozen-Π. Holding sectoral sorting fixed, the transfer relaxes the budget of unemployed and low-income informal households, and, since these are high-MPC, it dampens the aggregate response (Auclert, Rognlie, and Straub 2024).

The composition channel is captured by the moving-Π Jacobian, because the CCT raises the value of informality relative to formal work, so workers are tilted toward the informal sector, which is riskier. The insurance term is stabilizing, while the composition is potentially destabilizing. The sign and magnitude of the net effect are the quantitative answer to our question.

Letting the transition respond, $d\Pi_t = \Pi_{\mathbf{V}}\, d\mathbf{V}_t$, yields the general equilibrium Jacobian $\mathbf{G}$. This dynamic is obtained as the dynamic analogue of the steady-state: given a shock path, we iterate so that households' sectoral choices are consistent with the values those very choices generate along the transition (Auclert, Bardóczy, et al. 2021).

Calibration

Each model period stands for one quarter and the computational time horizon is 100 quarters (or 25 years). We solve the steady-state equilibrium computationally by Brent's method, internally calibrating parameters to clear the asset market, the labor market, the government budget constraint, and the Phillips curves. The external calibration is shown in table 3 and the endogenous calibration is in 4. The goods market was left untargeted to verify the Walras' Law.

The idiosyncratic productivity follows a log-normal $AR(1)$ process, $\ln e' = \rho_e \ln e + \varepsilon^e$, with $\varepsilon^e \sim N(0, \sigma_e^2)$. This is approximated by a finite Markov chain with $N_e = 15$ grid points (Rouwenhorst 1995). We also assume a log-spaced grid for assets $\ln a \in [\ln \underline{a}, \ln \bar{a}]$

Table 3: External Calibration

	Description	Value	Source
Households and Labor Market			
γ	EIS	0.5	Standard in Literature
δ_F	Job-Loss Probability to Formal	0.013	PNAD-C Data (Osório 2022)
δ_I	Job-Loss Probability to Informal	0.028	PNAD-C Data (Osório 2022)
σ_S	Smoothness of Labor Decision	0.5	Arbitrary
Productivity and Asset Grid			
$(\underline{a}, \bar{a})$	Log-Asset Grid	$(0, 200)$	No Borrowing
N_a	Number of Asset Grid	200	Accuracy near Constraint
ρ_e	Persistence of Log-Productivity	0.966	Kaplan, Moll, and Violante (2018)
σ_e	Productivity Std. Dev.	0.7	Arbitrary
μ_F	Formal Average Productivity	-0.219	PNAD-C Data
σ_F	Formal Std. Dev. Productivity	0.661	PNAD-C Data
σ_I	Informal Std. Dev. Productivity	0.972	PNAD-C Data
Firms			
Y	Aggregate Output	1.0	Normalization
μ, μ_w	Price Markup	1.11	Basu and Fernald (1997)
κ	Price-NKPC Slope	0.025	Hazell et al. (2022)
κ_w	Wage-NKPC slope	0.025	Equal to κ
Government and Monetary Policy			
r^*	Neutral Real Interest Rate	1.0%	4% annual average
ϕ	Taylor Rule Coefficient	1.5	Standard (Taylor 1993)
ϕ_B	Sensitivity to pay the Debt	0.2	Sufficient to B converges
τ_ℓ	Labor Income Tax Rate	0.2	Average Tax Payroll
T/w	Bolsa Família Transfer	0.1547	CadÚnico, PNAD-C

Table 4: Endogenous Calibration

	Description	Value	Target	Model	Data	Source
Households						
ψ	Disutility of Labor	0.236	h_F	1.0	1.0	Normalization
φ	Frisch Elasticity	0.200	X	X	X	PNAD-C Data
β_H	Patient's Discount Factor	0.963	B/Y	3.2	3.2	Tesouro Nacional
δ_β	Difference: $\beta_H - \beta_L$	0.12	X	X	X	Arbitrary
ω_I	Share of Impatient Agents	0.75	X	X	X	Arbitrary
Labor Market						
π_F	Formal Offer Probability	0.151	F	49.9%	49.8%	PNAD-C Data
π_I	Informal Offer Probability	0.269	I	44.5%	44.5%	PNAD-C Data
μ_I	Informal Average Productivity	-1.052	ξ	X	0.656	PNAD-C Data
$\bar{y}$	Bolsa Família Threshold	0.3	BF	14.8%	0.1611	CadÚnico

with $N_a = 200$ points, and a log-normal sector productivity grid $\ln \theta^s \sim N(\mu_s, \sigma_s^2)$, that is discretized by a Gauss-Hermite quadrature with $N_\theta = 5$ nodes (Meghir, Narita, and Robin 2015).

Each household has a $q = 0.1$ probability of drawing a new discount factor (β_L, β_H), corresponding to an average type duration of 25 years, used to capture a generation (Auclert, Rognlie, and Straub 2025). This aims to capture similar results in relation to a life-cycle overlapping generations with permanent ability shock model.

In the household preferences, elasticity of intertemporal substitution is standard in literature $\gamma = 0.5$[5]. The disutility of labor $\psi = 0.236$ is internally calibrated to normalize formal hours $h_F = 1.0$. The patient's discount factor $\beta_H = 0.963$ balances the asset market. We will use $\delta_\beta = 0.12$ and $\omega_I = 0.75$ to approximate the Wealth Gini Coefficient.

The layoff probabilities are equal to the transition probabilities in PNAD-C, $\delta_F = \mathbb{P}[\mathrm{F} \to \mathrm{U}] = 0.013$ and $\delta_I = \mathbb{P}[\mathrm{I} \to \mathrm{U}] = 0.028$. Further, the job offer probabilities are calibrated to match sector sizes $s \in \{F, I, U\}$ in equilibrium, as shown in Table 5. The other transition probabilities were left untargeted. The taste preferences smoothness is governed by $\sigma_S = 0.5$.

Table 5: Quarterly Labor-Market Transitions (%)

	Model			**PNAD-C**		
Origin	F	I	U	F	I	U
Formal	93.21	5.49	1.29	90.17	8.53	1.29
Informal	6.04	91.13	2.82	11.59	85.58	2.82
Unemployed	12.28	21.18	66.54	18.69	35.23	46.08
Stationary Share	49.86	44.46	5.68	49.84	44.52	5.63

We estimate the distribution of θ by fitting a log-normal to the observed wage distribution via maximum likelihood, which yields $\sigma_F = 0.661$ and $\sigma_I = 0.972$. We normalize average formal productivity to $\mathbb{E}(\theta^F) = 1$, so that $\mu_F = -0.219$. Finally, we set $\mu_I = -1.052$ to match the observed cross-sector wage gap $\xi := \mathbb{E}(y^F)/\mathbb{E}(y^I) = 0.656$.

The output is normalized $Y = 1.0$ and we defined $\mu = 1.11$, which is standard in the HANK literature (Kaplan, Moll, and Violante 2018; Auclert, Rognlie, and Straub 2025)[6], and $\kappa = \kappa_w = 0.025$, consistent with Hazell et al. (2022) estimates.

Taylor Rule is pretty standard, with $\phi = 1.5$ and $r^* = 1.0\%$, consistent with a 4% annual real interest rate for Brazilian economy (Santos and Muinhos 2024). The Government debt is set to 80% of yearly GDP, consistent with Brazilian National Treasure data. Wage tax is an average of all contributions levied on the payroll, $\tau_\ell = 0.2$.

The *Bolsa Família* statistics were obtained from the Cadastro Único[7], where we

[5]EIS $\gamma = 0.5$ is consistent with micro estimates surveyed by Valickova, Havranek, and Horvath (2015).
[6]Markup $\mu = 1.11$ is also consistent with the estimates of Basu and Fernald (1997).
[7]Available in VIS DATA 3 and Cadastro Único.

calibrated the T/w ratio in steady-state to be equal to the data from the Brazilian economy, with T = R\$ 670 and w = R\$ 4231. We also calibrated the *Bolsa Família* threshold $\bar{y} = 0.3$ to match the size of the program $BF = 0.1611$, considering only households in the labor force.

3. Results

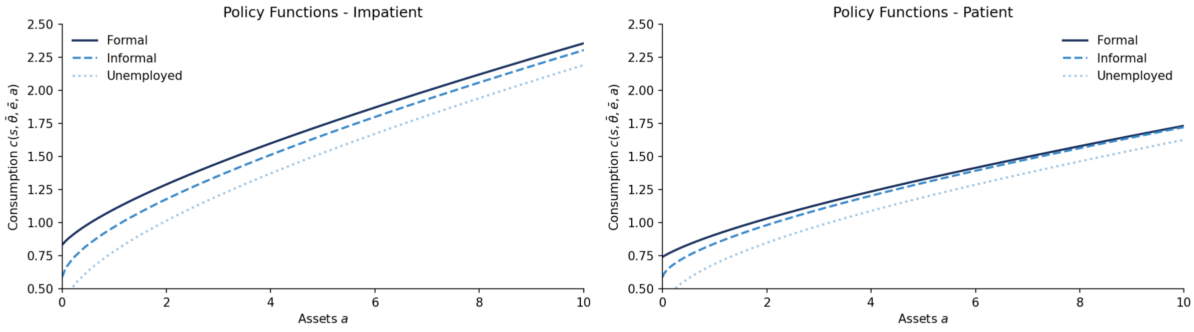

Figure 8: Steady-State Consumption Policy Function

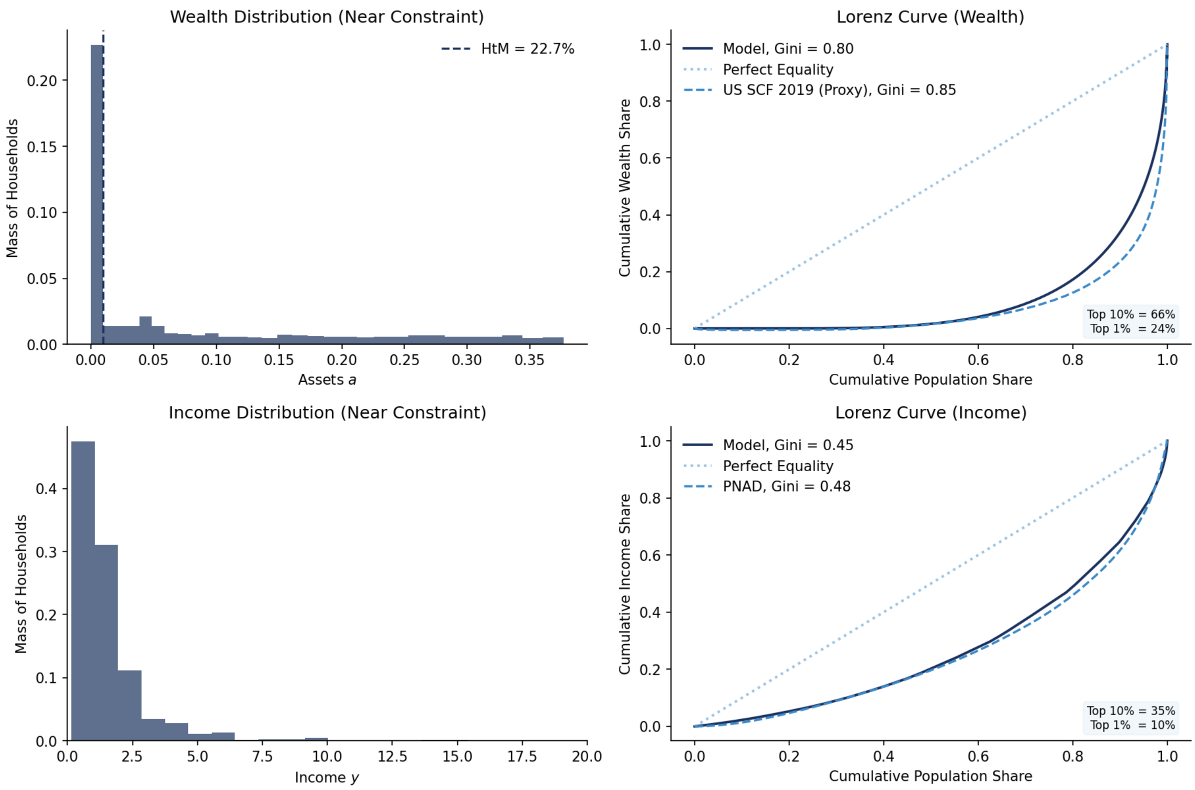

Figure 9: Wealth Distribution

Table 6: Steady State: Bolsa Família vs. No-Transfer Counterfactual

	With BF	No BF	Δ
Formal Share α_F	49.9%	63.8%	-14.0
Informal Share α_I	44.5%	31.1%	+13.3
Unemployment α_U	5.7%	5.0%	+0.7
Hand-to-Mouth	22.7%	12.6%	+10.1
Wealth Gini	0.799	0.764	+0.036
Consumption Gini	0.411	0.392	+0.019
Aggregate Consumption C	1.788	1.462	+0.327
BF Spending	0.023	0.000	+0.023
Welfare $\mathbb{E}[V]$	-8.627	-12.468	+3.840

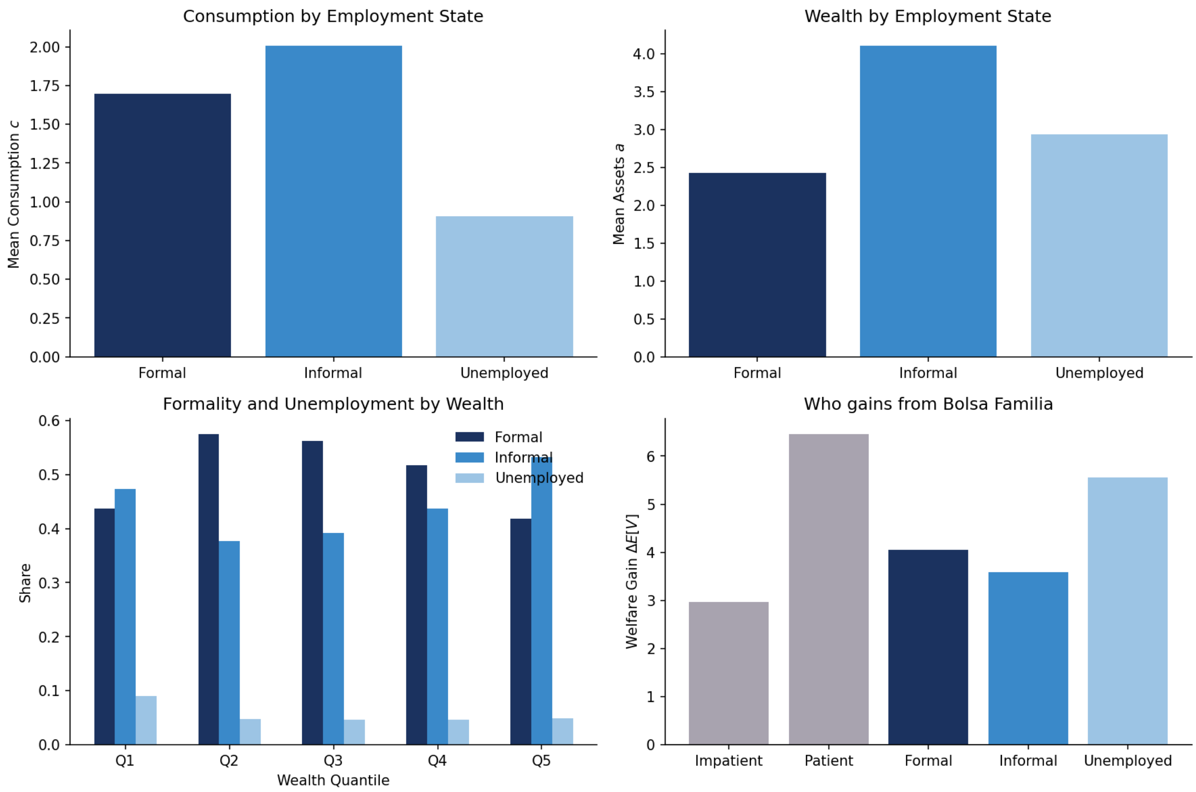

Figure 10: Steady-State Descriptive Statistics

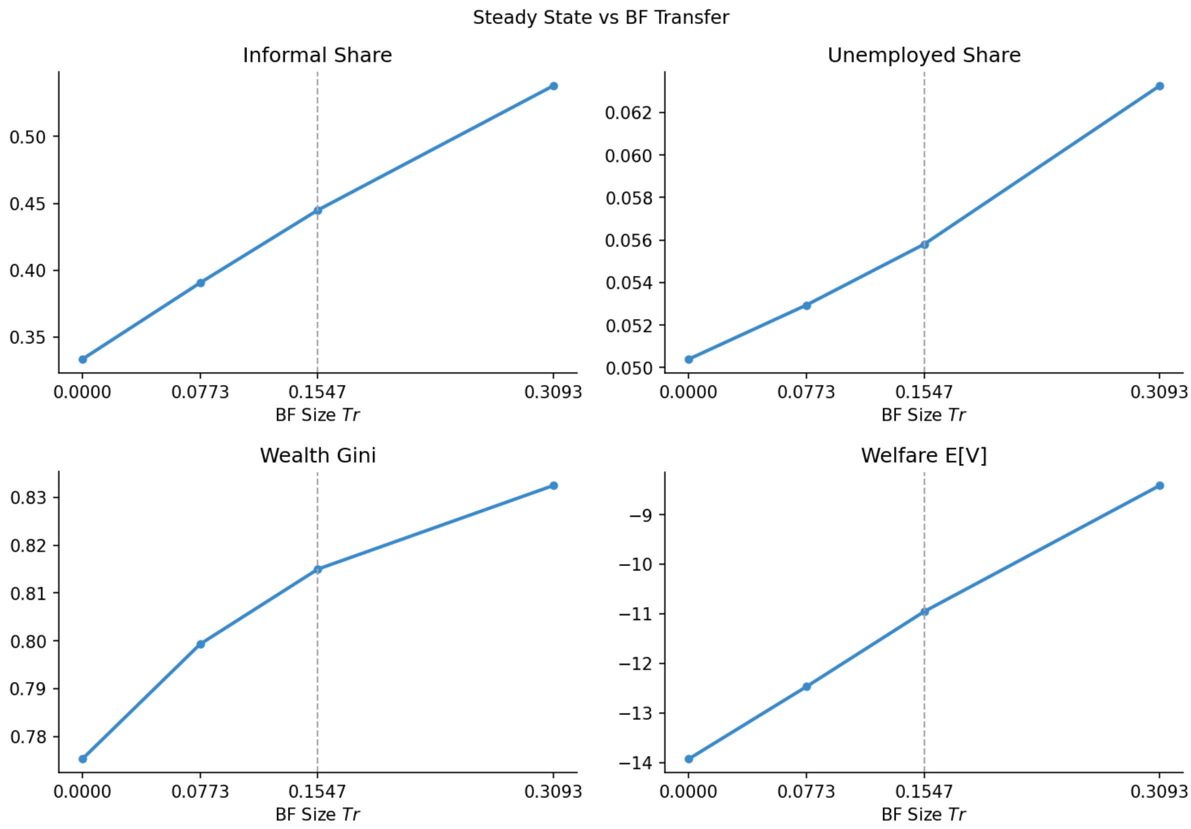

Figure 11: Steady-State Sensitivity Analysis

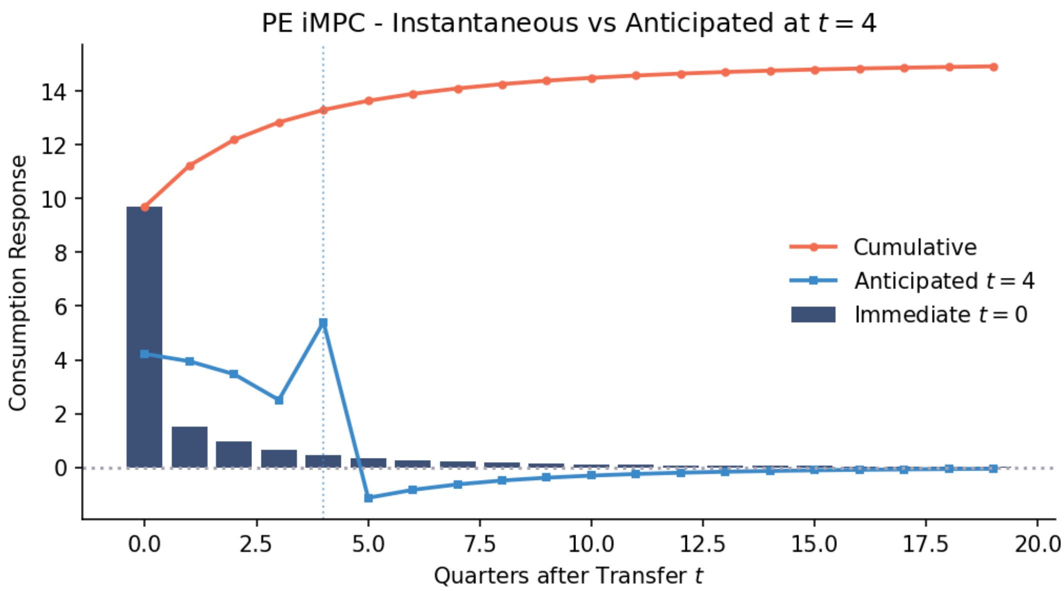

Figure 12: iMPC: Partial Equilibrium Profile

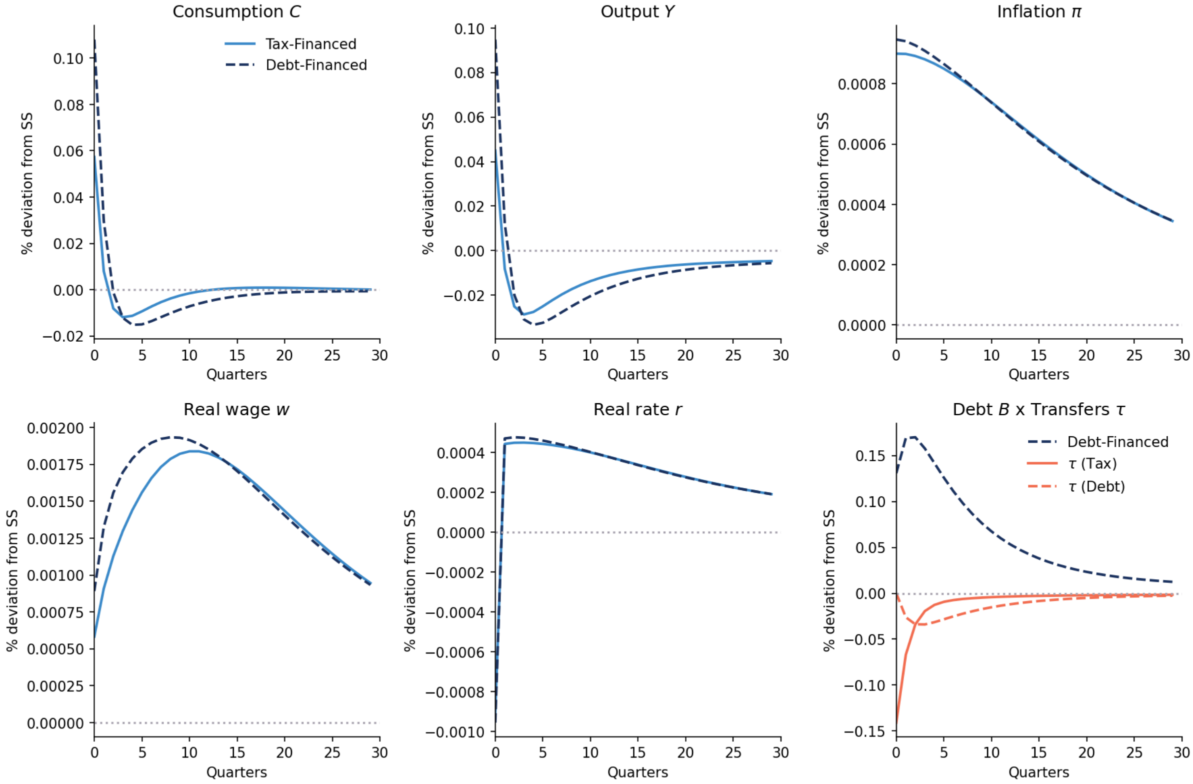

Figure 13: Response to a Shock in T by Financing Rules

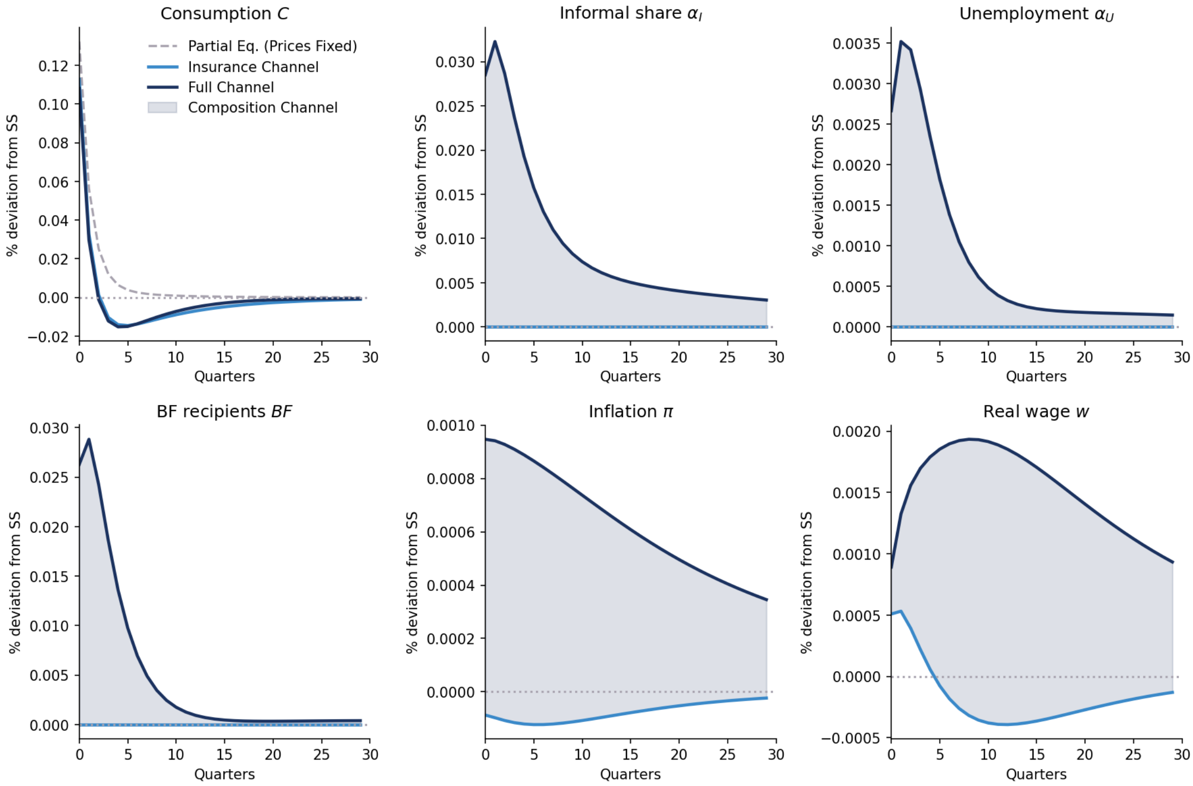

Figure 14: Response to a Shock in T by Channel

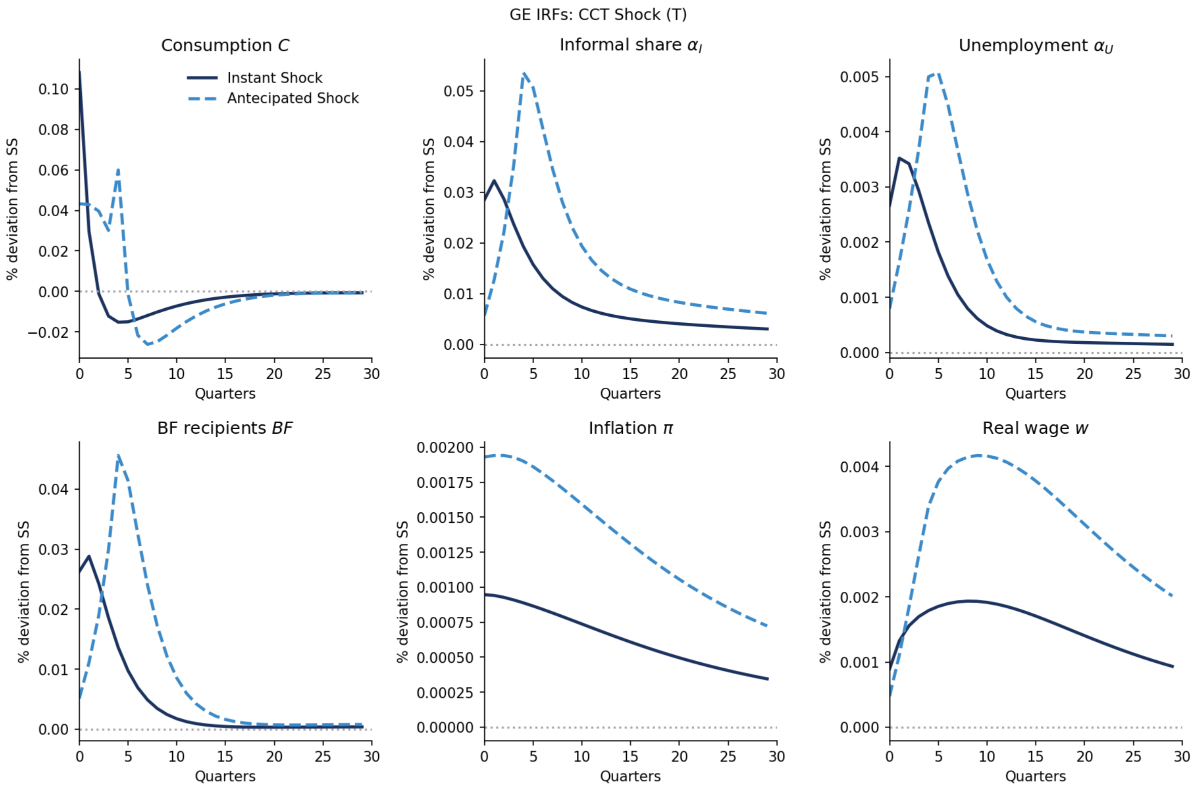

Figure 15: Response to a Shock in T: Instant vs Antecipated

Table 7: Cumulative Response (%), Shock in T

Variable	Horizon	Insurance	Full	Composition
Consumption C	1	0.113	0.108	-0.005
	4	0.137	0.124	-0.012
	20	0.005	0.013	0.008
	100	-0.022	-0.025	-0.003
Unemployment α_U	1	0.000	0.003	0.003
	4	-0.000	0.013	0.013
	20	-0.000	0.023	0.023
	100	-0.000	0.029	0.029
Inflation π	1	-0.000	0.001	0.001
	4	-0.000	0.004	0.004
	20	-0.002	0.015	0.017
	100	-0.003	0.028	0.030
Real wage w	1	0.001	0.001	0.000
	4	0.002	0.005	0.004
	20	-0.003	0.034	0.037
	100	-0.006	0.069	0.075

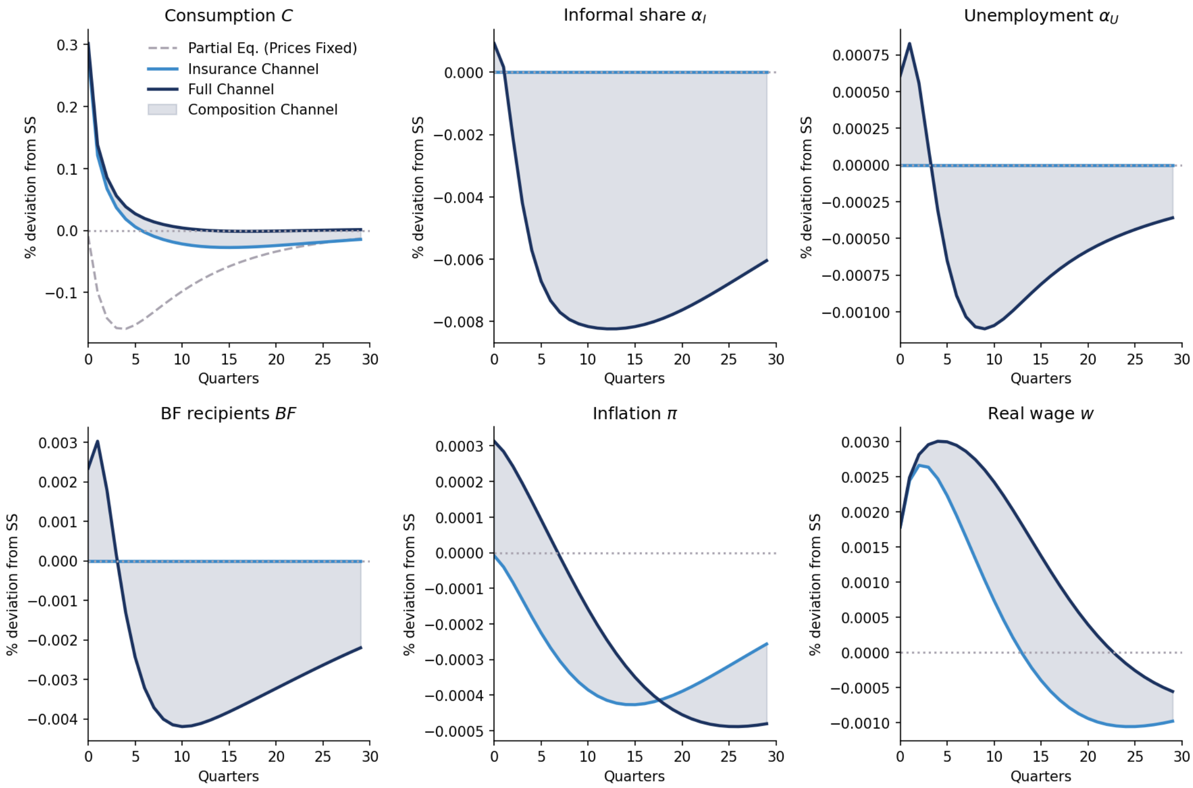

Figure 16: Response to an increase in i by Channel

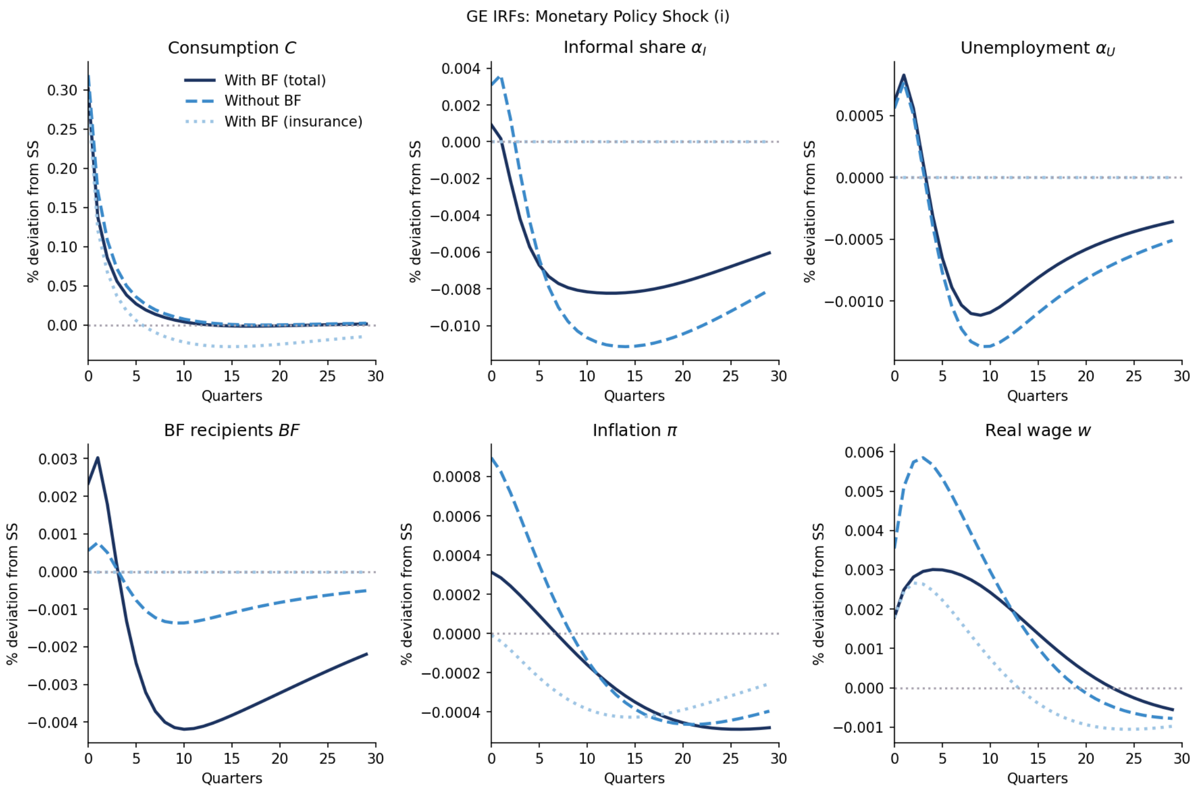

Figure 17: Response to an increase in i: With and Without CCT

Table 8: Cumulative Response (%), Shock in i

Variable	Horizon	Insurance	Full	Composition
Consumption C	1	0.288	0.301	0.014
	4	0.513	0.581	0.068
	20	0.234	0.697	0.463
	100	-0.129	0.779	0.908
Unemployment α_U	1	0.000	0.001	0.001
	4	-0.000	0.002	0.002
	20	-0.000	-0.011	-0.011
	100	-0.000	-0.024	-0.024
Inflation π	1	-0.000	0.000	0.000
	4	-0.000	0.001	0.001
	20	-0.006	-0.002	0.004
	100	-0.013	-0.022	-0.009
Real wage w	1	0.002	0.002	0.000
	4	0.010	0.010	0.001
	20	0.018	0.042	0.024
	100	-0.008	0.005	0.013

Robustness Tests

Lastly, in order to verify whether this results are representatives and robust, we intent to do some robustness tests changing some calibration parameters, such as γ, μ, and κ_w. This parameters might have minor implications in the results, but it is expected that the model continues to have the same results.

Also, we intent to perform other dividend specifications instead of $d_{i,t} = d_t \cdot e_{i,t}$. We also aim to change the GHH utility to a standard separable utility in the form of $u(c) - v(h)$, which must have similar results.

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Appendix

Price Phillips Curve

Let Y_t be the aggregate output of the economy, a CES of the intermediary inputs,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

This firm maximizes its output subject to $\int_0^1 p_{i,j} y_{i,j} \, dj$. The FOC implies that the demand for the intermediate good is given by

$$y_{j,t} = \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t,$$

where P_t is the equilibrium price of the final good,

$$P_t = \left(\int_0^1 p_{j,t}^{1-\varepsilon} \, dj \right)^{\frac{1}{1-\varepsilon}}.$$

We assume a Rotemberg (1982) quadratic price adjustment cost given by

$$\frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t,$$

where $\theta_p = (\varepsilon - 1)/\kappa$ is the adjustment cost multiplier and $\bar{\Pi}$ the steady-state inflation.

Let $\Xi(y)$ be the intermediary firm's cost. Given last period's price $p_{j,t-1}$, the firm chooses this period's price $p_{j,t}$ to maximize the present discounted values of future profits,

$$V_t^{IF}(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} y_{j,t} - \Xi(y_{j,t}) - \frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}^{IF}(p_{j,t}).$$

We can replace the demand function into this problem, so

$$V_t(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t - \Xi(y_{j,t}) - \frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}(p_{j,t}).$$

The FOC with respect to $p_{j,t}$ is

$$(1-\varepsilon) \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} \frac{Y_t}{P_t} + \varepsilon mc_{j,t} - \theta_p \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right) \frac{Y_t}{p_{j,t}} + \frac{1}{1+r_{t+1}} V'_{t+1}(p_{j,t}) = 0,$$

where $mc_t = \partial \Xi / \partial y_{j,t}$, and the envelope condition gives us

$$V'_{t+1} = \theta_p \left(\frac{p_{j,t+1}}{p_{j,t}} - \bar{\Pi} \right) \frac{p_{j,t+1}}{p_{j,t}} \frac{Y_{t+1}}{p_{j,t}}.$$

Thus, taking $p_{j,t} = P_t$ in equilibrium,

$$\theta_p (\pi_t - \bar{\Pi}) \pi_t = \varepsilon \left(mc_t - \frac{\varepsilon - 1}{\varepsilon} \right) + \frac{1}{1+r_{t+1}} \theta_p (\pi_{t+1} - \bar{\Pi}) \pi_{t+1} \frac{Y_{t+1}}{Y_t},$$

where we define the firm's markup as $\mu = \frac{\varepsilon}{\varepsilon-1}$. Linearizing around the steady-state $\pi_t = \bar{\Pi}$,

$$\ln(1+\pi_t) = \frac{\varepsilon}{\theta_p}\left(mc_t - \frac{1}{\mu}\right) + \frac{1}{1+r_{t+1}}\ln(1+\pi_{t+1})\frac{Y_{t+1}}{Y_t},$$

where $\kappa = \varepsilon/\theta_p$ and $mc_t = w_t/Z_t$. This environment is equivalent to Calvo (1983) in first order around zero inflation.

Wage Phillips Curve

We will use the same setting as Hagedorn, Manovskii, and Mitman (2019), but considering just the formal sector. Assume there is a union that aggregate effective labor input into

$$h_t = \left(\int_0^1 e_{i,t}(h_{i,t})^{\frac{\varepsilon_w-1}{\varepsilon_w}}\,di\right)^{\frac{\varepsilon_w}{\varepsilon_w-1}},$$

sells households labor services $e_{i,t}h_{i,t}$ at nominal wage $W_{i,t}$ to the labor recruiter, which minimizes costs $\int_0^1 W_{i,t}e_{i,t}h_{i,t}\,di$, implying a demand of

$$h_{i,t} = \left(\frac{W_{i,t}}{W_t}\right)^{-\varepsilon_w} h_t.$$

The unions sets a nominal wage $\hat{W}_t$ for an effect unit of labor, $W_{i,t} = \hat{W}_t$ to maximize profits subject to Rotemberg (1982)'s wage adjustment costs

$$e_{i,t}\frac{\theta_w}{2}\left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w\right)^2 h_t.$$

We assume that the union's wage setting problem is

$$V_t^w = \max_{\hat{W}_t}\int\left[\frac{(1-\tau_\ell)\hat{W}_t}{P_t}e_{i,t}h_{i,t} - \frac{v(h_{i,t})}{u'(\tilde{C}_t)} - e_{i,t}h_t\frac{\theta_w}{2}\left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w\right)^2\right]d\Lambda + \bar{\beta}V_{t+1}^w.$$

Using the same algebra as the price Phillips curve, assuming that the union allocates the same amount of hours in each worker $h_{i,t} = h_t$, and using $\int_F e_{i,t}d\Lambda = L_t$, it is possible to find that

$$\theta_w(\pi_t^w - \bar{\Pi}^w)\pi_t^w = \varepsilon_w\left[\frac{v'(h_t)}{u'(\tilde{C}_t)} - \frac{\varepsilon_w - 1}{\varepsilon_w}(1-\tau_\ell)w_tL_t\right] + \bar{\beta}\theta_w(\pi_{t+1}^w - \bar{\Pi}^w)\pi_{t+1}^w\frac{h_{t+1}}{h_t},$$

which we can linearize around the steady state $\pi_t^w = \bar{\Pi}^w$ and take $h_{t+1} = h_t$, since there is no population nor technological growth. Thus,

$$\ln(1+\pi_t^w) = \frac{\varepsilon_w}{\theta_w}\left[\frac{v'(h_t)}{u'(\tilde{C}_t)} - (1-\tau_\ell)w_tL_t\frac{\varepsilon_w - 1}{\varepsilon_w}\right] + \bar{\beta}\ln(1+\pi_{t+1}^w).$$

Therefore, defining $\kappa_w = \varepsilon_w/\theta_w$ and $\mu_w = \varepsilon_w/(\varepsilon_w - 1)$, the wage Phillips curve is

$$\ln(1+\pi_t^w) = \kappa_w\left[\psi\cdot h_t^{1/\varphi}\cdot\tilde{C}_t^{1/\gamma} - \frac{(1-\tau_\ell)w_tL_t}{\mu_w}\right] + \bar{\beta}\ln(1+\pi_{t+1}^w).$$